

Quasi-random oriented molecular contacts for inverted perovskite solar cells with improved efficiency

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Tianyu Li^{1,2,10}, Wenlin Jiang^{3,10}, Tonghui Wang^{4,10}, Yingguo Yang⁵, Jiali Liu⁶, Yaxin Zhai⁶, Chenkai Liu^{1,2}, Miaoling Lin^{1,2}, Pingheng Tan^{1,2}, Zhenwei Jiang⁷, Xinpeng Wang⁷, Hengyu Zhang⁸, Zhenyi Ni⁸, Qingqing Dai⁹, Chun-To Wong³, Zhaoyang Han^{1,2}, Zhenhan Wang^{1,2}, Xingwang Zhang^{1,2}, Qing Jiang⁴, Alex K.-Y. Jen³✉ & Qi Jiang^{1,2}✉

Self-assembled hole-selective molecules (SHMs) can enhance the efficiency and stability of inverted perovskite solar cells (PSCs). Their molecular structures and assembly arrangement at buried interfaces determine charge-transfer dynamics, perovskite crystallization and photovoltaic performance. We design self-assembled molecules featuring a laterally extended π -scaffold by attaching two flanking phenyl groups onto the 7*H*-dibenzo[*c,g*]carbazole. This design manipulates the molecular packing, resulting in a quasi-random oriented assembly on the substrate to accelerate the interfacial hole-transfer kinetics at both the substrate/SHM and SHM/perovskite interfaces. The solar cells achieve a stabilized power conversion efficiency of 27.1% (certified stabilized 26.67%) for a small-area PSC and 26.0% (certified stabilized 25.94%) for a 1-cm² device. The small-area device retains 95% of its initial efficiency over 1,630 hours under 1-sun operation at 65 °C and 91% over 1,240 hours operation at 85 °C. These findings provide insights for designing improved self-assembled molecular contacts for inverted PSCs.

The promises and rapid advancement of metal halide perovskite research have driven the exploration of perovskite solar cells (PSCs) towards their theoretical solar energy conversion efficiency limits and the operational stability targeting outdoor applications^{1–4}. P–i–n structured (inverted) PSCs have emerged as particularly promising due to their scalability and compatibility with both single-junction and tandem configurations^{5–8}. A critical research focus in this device architecture currently involves the development of self-assembled hole-selective molecules (SHMs), which offer exceptional hole extraction capability, minimal charge transport resistance and thus reduced energy losses at the buried interface^{9–12}. Their easy synthesis, good processibility and favourable compatibility with flexible substrates and tandem fabrications to enable their commercial

viability, continuously motivating the researchers to expand the SHM library^{12–14}.

The tunable structures and functionalities of SHMs allow precise modulation of their chemical and electrical properties for achieving optimal buried interface between conducting substrate and perovskite layer^{15–18}. Recent advancements, including the improved interfacial energetic alignments and molecular dipole engineering to enhance hole-transfer, optimized molecular reorganization on oxide substrates for assisting perovskite deposition, and enhanced chemical and physical stability, have shown the great values in developing new SHMs^{19–22}. However, there is a significant disconnection between molecular design principles and the resultant film properties, particularly regarding their self-assembly behaviour and interfacial

A full list of affiliations appears at the end of the paper. ✉e-mail: alexjen@cityu.edu.hk; qjiang@semi.ac.cn

interactions with the overlying perovskite layer. Among the developing molecules, (4-(7*H*-dibenzo[*c,g*]carbazole-7-yl)butyl)phosphonic acid (CbzNaph) has been standing out to behave effectively in various high-performance PSC architectures^{10,20,23,24}. It features a helical π -extended carbazole skeleton, with improved molecular dipole moment and strengthened π - π interaction, alongside ordered assembly²³, while its single conformation structure formed in the one-dimensional linear assembly creates difficulty to form perfect large-area uniform film on substrate. Therefore, it is important to introduce more interactive modes to mitigate this dilemma and dig out more universal principles to guide the further progresses.

In this work, we present a rational designed SHMs via the twisted π extension strategy. By further laterally extending two phenyl groups at the 5,9-positions of the π -conjugated framework of the reference CbzNaph, we constructed the corresponding (4-(5,9-diphenyl-7*H*-dibenzo[*c,g*]carbazole-7-yl)butyl)phosphonic acid, termed as 2Ph-CbzNaph. Both theoretical and experimental analyses validate the fascinating traits of the structure: (1) the intermolecular electronic coupling was strengthened through the twisted phenyl side groups to facilitate intermolecular hole transfer; (2) it resulted in increased random molecular motion and provided conformational flexibility and then quasi-random molecular arrangement on substrates, which enhanced the statistically probability of achieving optimal interfacial charge-transfer kinetics. Moreover, the derived uniform molecular stacking provided a favourable template for perovskite growth to minimize trap-assisted non-radiative recombination at the buried interface. As a result, PSCs incorporating 2Ph-CbzNaph achieved outstanding stabilized power conversion efficiency of 27.1% (certified stabilized 26.67%) for a 0.0725 cm² cell and 26.1% (certified stabilized 25.94%) for a 1 cm² cell. Notably, the target device retained over 90% of its original efficiency after over 1,240 hours of thermal-photo operation under continuous 1-sun soaking at 85 °C. These results highlight the potential of employing SHM structural engineering for achieving high-efficiency and stable perovskite photovoltaics.

Molecule design and properties

The synthetic route for 2Ph-CbzNaph is shown in Supplementary Note 1, along with structural characterizations presented in Supplementary Figs. 1–9. The molecular structure is shown in Fig. 1a. This modification achieves two key objectives: (1) enhancing intermolecular interactions through horizontal molecular extension (Supplementary Fig. 13) and (2) creating significant steric hindrance between the phenyl unit and skeletal hydrogen atoms to form large torsion angles of 45.4° and 52.4° relative to the π -scaffold (Supplementary Fig. 14). The resulting expanded rigid unit space volume provides a larger tolerance space for compensating the wagging of flexible alkyl linker and more regulated molecular stacking of 2Ph-CbzNaph to achieve uniform distribution on the substrate. Note that the colloidal sizes of our SHMs in processing solution of ethanol showed no discernible difference, as evidenced by the dynamic light scattering results (Supplementary Fig. 15).

The crystallization behaviour and molecular orientation were first investigated by grazing-incidence wide-angle X-ray scattering (GIWAXS) to understand the impacts of the phenyl substituents. As shown in Fig. 1b,c, the two-dimensional (2D) GIWAXS patterns revealed obvious differences between the two SHMs. The control CbzNaph sample exhibited discrete Bragg spots in its diffraction pattern, indicating high crystallinity with preferred molecular orientations and well-organized packing. In contrast, the 2Ph-CbzNaph sample displayed the characteristic polycrystalline Debye–Scherrer rings, suggesting a less-preferred oriented growth. As for angle-resolved polarized Raman (ARPR) measurements^{25,26} (Fig. 1d,e, Supplementary Note 2 and Supplementary Fig. 16), the Raman peak centred at ~1,600 cm⁻¹ (assigned to the C=C stretching vibration of the π -conjugated backbone) for CbzNaph²⁷, which exhibited pronounced anisotropy and

could be roughly fitted by a $\cos^4\theta$ function (magenta line in Fig. 1d). This is the characteristic of a crystalline structure with preferential molecular alignment²⁶. While 2Ph-CbzNaph displayed weaker polarization dependence, suggesting a relatively random oriented arrangement. We further elucidated the surface conformation of the SHMs, by using highly interface-sensitive sum-frequency generation vibrational spectroscopy (SFG-VS)²⁸ (Supplementary Note 2). As shown in Fig. 1f, CbzNaph exhibited a more pronounced peak at ~3,063 cm⁻¹ (assigned to the aromatic C–H stretching mode) in the *ssp* SFG spectra²⁰, compared with 2Ph-CbzNaph. This unveiled that CbzNaph adopts a predominantly ordered arrangement along the surface normal, whereas 2Ph-CbzNaph shows a less-preferred oriented configuration. Quantitative analysis yielded average tilt angles of 25° for the conjugated core of CbzNaph and 37° for 2Ph-CbzNaph (Supplementary Fig. 18).

X-ray photoelectron spectroscopy (XPS) measurements (Supplementary Fig. 20) semi-quantitative provided the coverage information by revealing stronger phosphorus (P) 2*p* signal absorbed on the substrate for 2Ph-CbzNaph compared to control. Concurrently, there was reduced intensities of tin (Sn) and oxygen (O) signals from the underlying fluorine-doped tin oxide (FTO) substrate, and the P/Sn ratio on 2Ph-CbzNaph-coated substrate was found to be 0.096, which is higher than that of CbzNaph counterpart of 0.069. These confirmed the superior surface coverage achieved by 2Ph-CbzNaph. Next, we determined the surface density of SHMs coated on FTO by cyclic voltammetry (CV)²⁹ (Supplementary Fig. 21). The surface density showed an increase from 3.8 × 10¹³ for CbzNaph to 5.1 × 10¹³ molecules cm⁻² for 2Ph-CbzNaph, thereby confirming its denser surface coverage.

To gain insight into the spatial distributions, we performed independent molecular dynamics (MD) simulations. There was assembly difference that molecular aggregate formation was observed in the case of CbzNaph, whereas 2Ph-CbzNaph exhibited a more homogeneous distribution (Fig. 1g,h and Supplementary Fig. 22). Kelvin probe force microscopy (KPFM) measurements and contact potential difference analysis also showed that 2Ph-CbzNaph covered FTO possesses more homogeneous and concentrated potential distribution with a narrower full width at half maximum (Fig. 1i,j and Supplementary Figs. 23 and 24), compared to the non-uniform potential distribution observed for the CbzNaph control.

As for lateral intermolecular charge transfer, based on the Marcus electron transfer theory, three key determinants are essential, including the reorganization energy (λ), driving force (ΔE) and transfer integral (v) controlling the hopping rate³⁰, where λ represents the energy cost associated with the rearrangement of the atomic nuclei after an electron has hopped from one site to another; ΔE represents the thermodynamic preference for the electron transfer to occur; v represents the coupling strength due to orbital overlaps. There are two density functional theory (DFT) optimized configurations (denoted as configuration 1 for ideal face-to-face and configuration 2 for in-plane displacement) for SHM dimer anchoring on tin oxide (SnO₂) (110) plane (Fig. 1k and Supplementary Fig. 25). With configuration 1 in Fig. 1k,l, 2Ph-CbzNaph exhibited a substantial hole-transfer rate (k) augmentation (1.32 × 10¹³ s⁻¹ for 2Ph-CbzNaph versus 2.23 × 10¹¹ s⁻¹ for CbzNaph) due to smaller λ (211 meV), larger ΔE (30 meV) and v (40 meV) compared to the control (316 meV, 21 meV and 10 meV, respectively), which was the same trend in configuration 2 (Supplementary Figs. 25 and 26). These characteristics suggest the enhanced lateral intermolecular interactions, which could facilitate achieving the molecule coverage uniformity and overall charge transport. In parallel, we also performed conductive atomic force microscopy (C-AFM) to evaluate the conductance of SHMs (Fig. 1m,n). The mean current of 2Ph-CbzNaph is found to be 1.54 nA, which is twice that of CbzNaph (0.69 nA), indicating its feasibility to mitigate the transport loss. This scenario was then corroborated by space-charge-limited current measurements (Supplementary Fig. 27a), which delivered a hole mobility of 1.1 × 10⁻⁶ cm² V⁻¹ s⁻¹ for 2Ph-CbzNaph, higher than that of CbzNaph with 6.8 × 10⁻⁷ cm² V⁻¹ s⁻¹. Moreover, conductivity measurement further

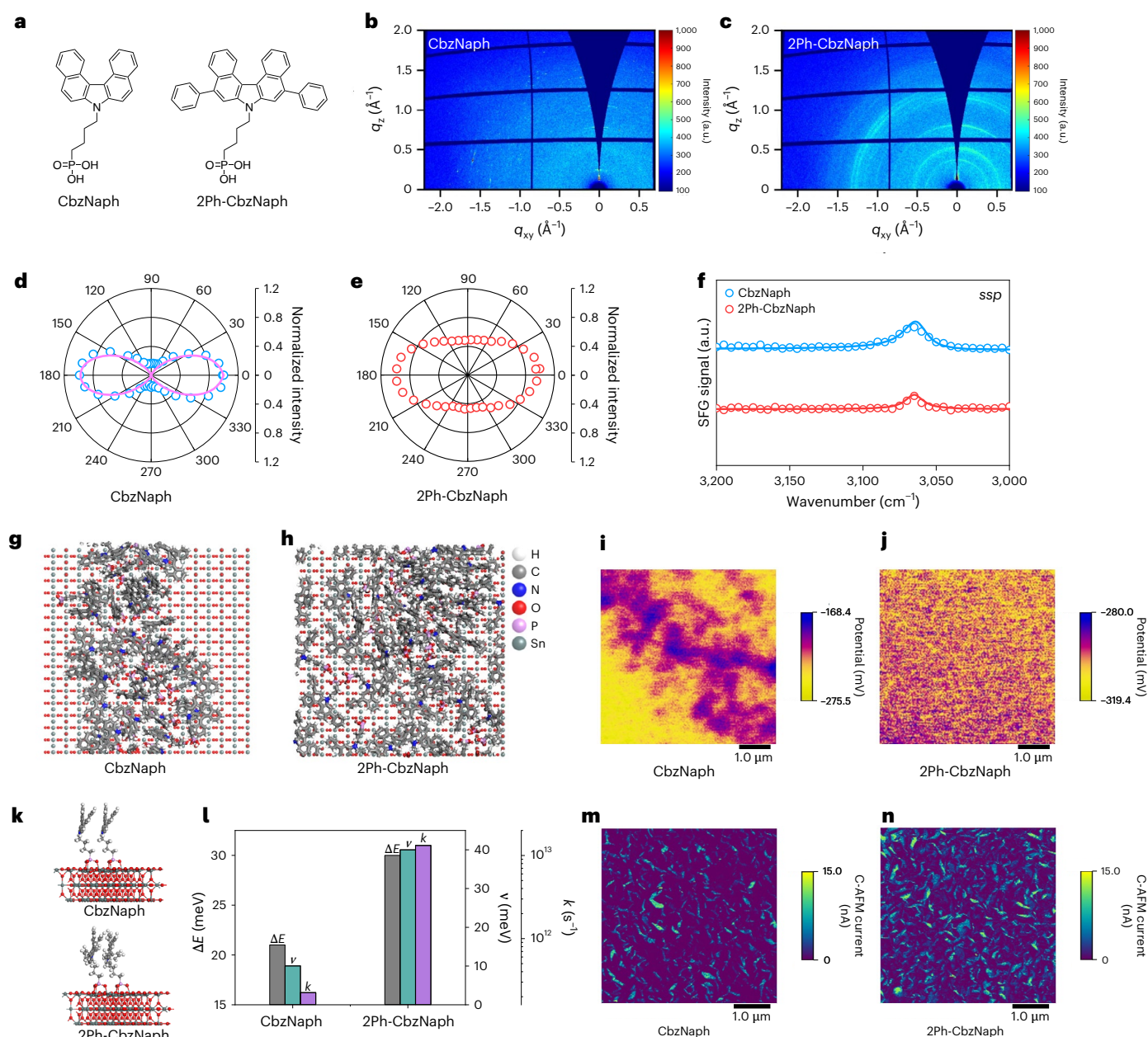


Fig. 1 | Comparisons of molecular properties on substrates. **a**, Chemical structures of two SHMs of CbzNaph and 2Ph-CbzNaph. **b, c**, Grazing-incidence wide-angle X-ray scattering patterns of the two SHMs (CbzNaph (**b**) and 2Ph-CbzNaph (**c**)). **d, e**, ARPR intensity profiles of CbzNaph (**d**) and 2Ph-CbzNaph (**e**) films. The magenta curve in **d** represents a fit to the function $y = \cos^4\theta$, an indicative of preferential molecular alignment of CbzNaph. **f**, Sum-frequency generation vibrational spectra with *ssp* polarization. The data were fitted via the Lorentzian equation, detailed in Supplementary Note 2. **g, h**, Top views of the equilibrated molecular representations (CbzNaph (**g**) and 2Ph-CbzNaph (**h**)) on

the surface of SnO₂ examined by MD simulations. **i, j**, Surface contact potential difference images obtained by Kelvin probe force microscopy of SHM-coated substrates (CbzNaph (**i**) and 2Ph-CbzNaph (**j**)). **k, l**, Intermolecular charge transfer among neighbouring molecules by DFT calculations, showing the representative binding of ideal face-to-face of the two SHMs on SnO₂ (**k**) and calculated key parameters, including driving force (ΔE), transfer integral (ν) and the derived hole-transfer rate (k) comparisons (**l**). Another representative binding with in-plane displacement was provided in Supplementary Figs. 25 and 26. **m, n**, C-AFM images of CbzNaph (**m**) and 2Ph-CbzNaph (**n**).

confirmed improved charge transport, yielding values of $1.02 \mu\text{S cm}^{-1}$ for 2Ph-CbzNaph versus $0.48 \mu\text{S cm}^{-1}$ for CbzNaph (Supplementary Fig. 27b). The overall improved charge transport behaviour should be beneficial to buried interface quality and device performance.

Buried interface charge-transfer dynamics analysis

To evaluate the potential of these two SHMs in serving as an efficient hole-selective layer at the buried interfaces, we systematically compared their molecular orientation distributions and charge-transfer

properties on conductive substrates. From the MD simulation results, the orientation angular distributions (Fig. 2a) revealed that 2Ph-CbzNaph possess a more broadly distributed arrangement, as quantified by the derived higher Shannon entropy (H), which means a higher possibility of orderless (Supplementary Note 3). The trend is consistent when SHMs are randomly distributed on substrate surface initially (Supplementary Note 3 and Supplementary Fig. 28).

To investigate the charge-transfer properties between the SHMs and the substrate in terms of the orientation, we performed DFT calculations on the charge-transfer potential between SHMs and conducting

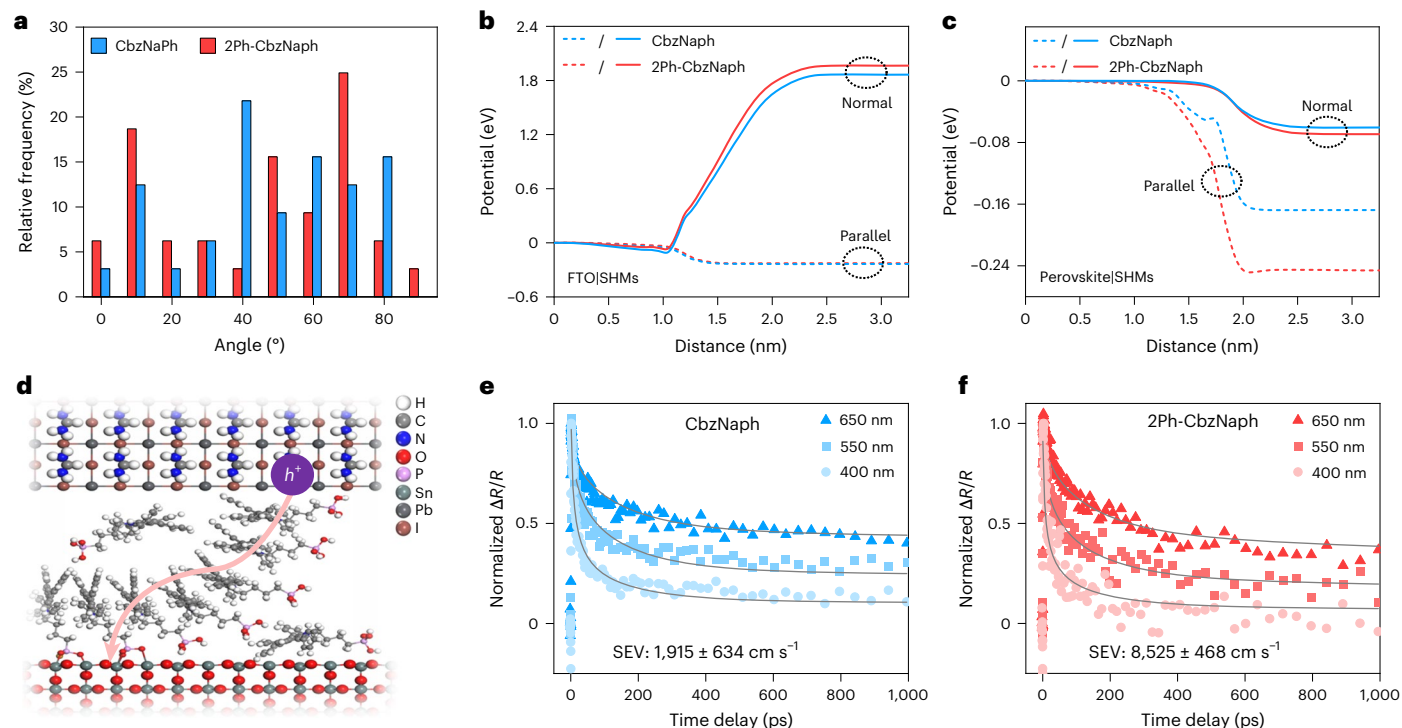


Fig. 2 | Molecule arrangements at buried interface and charge-transfer properties. **a**, Orientation angular distributions extracted from MD simulations. **b,c**, Calculated charge-transfer potential line profile along the vertical direction relative to the tin oxide surface (**b**) and perovskite surface (**c**) when SHMs absorbed with normal and parallel orientations. **d**, Proposed schematic illustration of the molecule contacts between conducting substrates and perovskites, depicting the idealized and optimal hole-transfer preferences

(pink arrow) related to molecule orientations. The purple circle represents photogenerated holes. **e,f**, The normalized surface carrier kinetics for perovskite films deposited on SHM-coated substrates (CbzNaph (**e**) and 2Ph-CbzNaph (**f**)) photoexcited at the pump energies of 3.1, 2.25 and 1.91 eV. The grey solid traces are nonlinear least-squares global best-fit curves via a diffusion-surface extraction model. SEV, surface extraction velocity.

substrate at two extreme conditions of normal and parallel (Fig. 2b and Supplementary Fig. 29). The results indicated that, in the normal orientation, both molecules exhibited favourable potential energy gradients for hole transfer to FTO, with 2Ph-CbzNaph showing a higher driving potential (1.96 eV versus 1.86 eV for CbzNaph). Conversely, the parallel orientation brought forth a comparable potential barrier (0.22 eV to 0.24 eV) impeding charge transfer. When the interactions between SHMs and perovskites were explored (Fig. 2c and Supplementary Fig. 30), the parallel orientation enables notable highest occupied molecular orbital delocalization across the interface, facilitating hole extraction, whereas the normal orientation leads to localized electron density primarily over the molecule itself. There is also noticeably larger driving force for better hole extraction in the case of 2Ph-CbzNaph under the parallel configuration, compared to CbzNaph (0.24 eV versus 0.17 eV). Therefore, the quasi-random distribution of 2Ph-CbzNaph better satisfies: (1) normal alignment of the SHMs relative to FTO surface facilitate optimal SHM-to-FTO hole injection and (2) parallel alignment relative to the perovskite surface benefits for efficient perovskite-to-SHM hole extraction. Therefore, we proposed a schematic diagram of the preferential hole transfer as Fig. 2d. It suggests that efficient overall hole transfer from perovskite to the FTO electrode requires a smooth orientation transition in SHM orientation from parallel (at the perovskite interface) to normal (at the FTO interface). This condition necessitates quasi-random oriented molecular contacts, which was observed in 2Ph-CbzNaph arrangement that it enhances the statistical probability of achieving optimal charge transfer. Note that our DFT calculations offer a plausible explanation that the quasi-random orientation conformation facilitates interface charge-transfer dynamics, while there might be other possible mechanisms that could plausibly account for the observed device performance improvements in the later sections.

The photophysical properties were analysed by fitting time-resolved photoluminescence (TRPL) decay through a biexponential

function and deriving differential lifetime, with the half-device sample of FTO/SHMs/perovskite (Supplementary Fig. 31). The faster initial decay and sharper rise at early times of differential lifetime indicate the more efficient interfacial charge-transfer process with 2Ph-CbzNaph³¹. And the slower decay of TRPL lifetime for target sample implies a lower recombination in the bulk and indicated a higher quality of the perovskite layer^{32,33}. For morphological study, the scanning electron microscopy (SEM) images revealed that there were no discernible differences in the morphology of the top surface perovskite films when depositing on CbzNaph and 2Ph-CbzNaph, both exhibiting micrometre-sized grains (Supplementary Fig. 32). However, the crystallinity of perovskite film was enhanced, as supported by the increased X-ray diffraction peak intensities and reduced full width at half maximum values (Supplementary Fig. 33), which is consistent with that obtained from the TRPL results. To gain more insight into the buried surface charge-transfer kinetics, we employed transient reflection spectroscopy study^{34,35}. 2Ph-CbzNaph exhibited obviously higher surface extraction velocity ($8,525 \pm 468 \text{ cm s}^{-1}$) than that of the CbzNaph ($1,915 \pm 634 \text{ cm s}^{-1}$) (Fig. 2e,f and Supplementary Fig. 34). This finding is also consistent with those from TRPL results.

The evaluation of the energy alignment was performed by ultra-violet photoelectron spectroscopy (UPS), as shown in Supplementary Figs. 35 and 36, the target SHM-coated FTO has a deeper work function, which agrees well with the trend obtained by KPFM measurements. In addition, the valence band maximum energy level (E_{VBM}) is estimated to be -5.35 eV compared with that of the control, -5.27 eV . Note that the thickness of the SHM layers were determined to be 2.5 nm and 2.7 nm for CbzNaph and 2Ph-CbzNaph by X-ray reflectivity measurement, respectively, through X-ray reflectivity (Supplementary Fig. 37). The derived UPS results should therefore predominantly originate from SHMs itself.

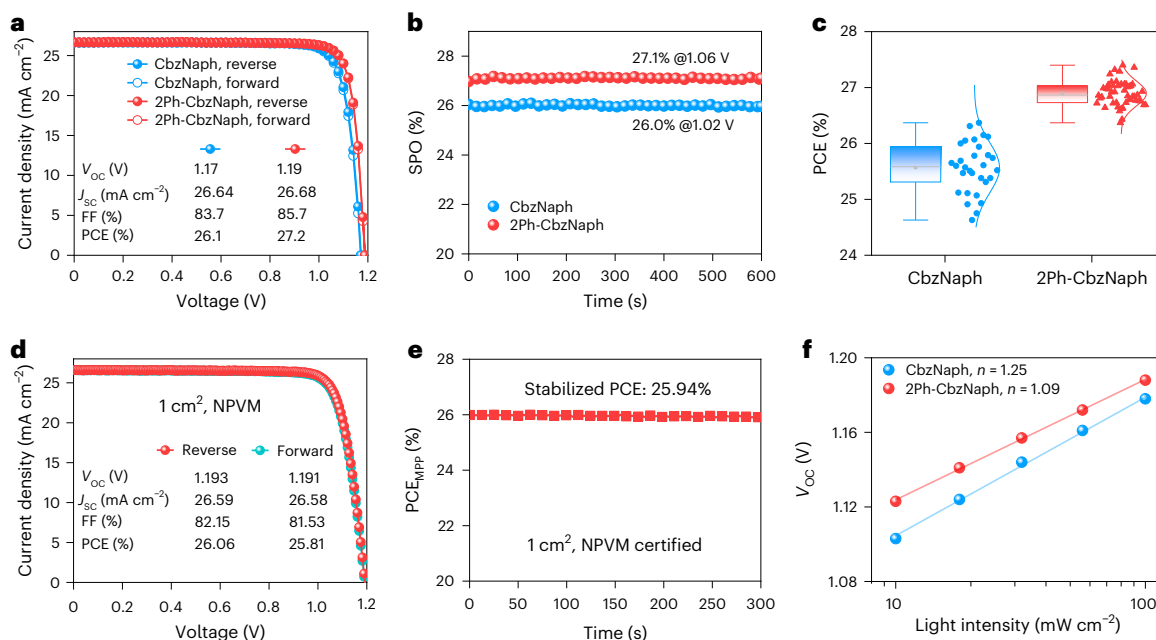


Fig. 3 | Device efficiency comparisons. **a**, Current density–voltage (J – V) characteristic of typical CbzNaph- and 2Ph-CbzNaph-based inverted PSCs under 100 mW cm⁻², AM 1.5 G illumination. Device structure: FTO/SHMs/perovskite/modification layer/C₆₀/atomic layer deposition (ALD)-SnO₂/Ag. **b**, The corresponding stabilized power output (SPO) efficiency under their maximum power point voltage, after tracking for 10 min. **c**, The device efficiency statistic data collection based on 30 CbzNaph and 50 2Ph-CbzNaph devices; the data are

reverse scan efficiency recorded under AM 1.5 G illumination in a nitrogen-filled glovebox. The box plots illustrated the mean, median line and 25–75% box limits with interquartile range whisker, with the data following a normal distribution. **d**, J – V curves of a 1 cm² 2Ph-CbzNaph-based device, measured by NPVM. **e**, The corresponding maximum power point tracking power conversion efficiency (PCE_{MPP}) under simulated AM 1.5 G sunlight in air. **f**, Light intensity dependence of V_{oc} . The solid lines indicated linear fits.

A better E_{VBM} alignment with the perovskite will help reduce interfacial recombination losses.

Photovoltaic efficiency measurements and characterizations

Building upon the promising results from the SHM characterizations, we fabricated inverted PSCs with the architecture of FTO/SHMs/Cs_{0.025}Rb_{0.025}FA_{0.95}PbI₃/modification layer/C₆₀/SnO₂/Ag (Supplementary Fig. 38) to further verify the effectiveness of 2Ph-CbzNaph in enhancing the efficiency of the derived devices. For the comparisons of optimal current density–voltage (J – V) testing between CbzNaph and 2Ph-CbzNaph, as depicted in Fig. 3a, the reference device exhibited a power conversion efficiency (PCE) of 26.1%, with an open-circuit voltage (V_{oc}) of 1.17 V, short-circuit current density (J_{sc}) of 26.64 mA cm⁻² and fill factor (FF) of 83.7% with minimal hysteresis. In contrast, the target device showed an elevated PCE of 27.2%, with V_{oc} of 1.19 V, J_{sc} of 26.68 mA cm⁻² and FF of 85.7% without hysteresis (Supplementary Table 2). The corresponding integrated J_{sc} values derived from external quantum efficiency (EQE) spectra were close as 26.07 mA cm⁻² and 26.08 mA cm⁻², matching well with the J – V measurements (Supplementary Fig. 39). The bandgap of the perovskite absorber derived from the EQE derivative was 1.52 eV. In addition, their stabilized power output efficiency is 26.0% (at 1.02 V) and 27.1% (at 1.06 V), respectively, after tracking for 600 s (Fig. 3b). The improved FF, V_{oc} and efficiency are probably due to better charge-transporting properties at the buried interface. It exhibited a $V_{oc} \times FF$ product beyond 91% of the Shockley–Queisser limit. The loss analysis of FF also showed the effectively attenuated non-radiative recombination loss at the buried interface of the target device (Supplementary Fig. 40). One of the target devices was validated by an independent authorized testing centre (National Photovoltaic Industry Measurement and Testing Center, NPVM) and exhibited a stabilized efficiency of 26.67% after 300 s maximum power point tracking (Supplementary Fig. 41). These results are among the highest

efficiencies ever reported (Supplementary Table 3). To compare the reproducibility, we collected 30 reference and 50 target devices, target device exhibited better average yields and narrower distributions in all FF, V_{oc} and overall PCE (Fig. 3c and Supplementary Fig. 42).

To demonstrate the general applicability of the designed SHMs, we also fabricated a 1-cm² aperture area device. It delivered a PCE of 26.06% under reverse scan with a V_{oc} of 1.193 V, a J_{sc} of 26.58 mA cm⁻² and an FF of 82.2%, along with the stabilized certified efficiency of 25.94%, validated by NPVM (Fig. 3d,e and Supplementary Fig. 43). This is among the highest certified PCEs reported for 1-cm² single-junction PSCs so far and is also consistent with our laboratory measurement results (Supplementary Fig. 44). The efficiency gap caused by scaling of area is significantly alleviated in our device system.

To validate the results of the improved photovoltaic performance, we also conducted light intensity dependent device J – V response study (Fig. 3f and Supplementary Fig. 45). The semi-logarithmic plot of V_{oc} versus light intensity underwent linear fitting, facilitating the derivation of the ideality factor (n) from the slope. This analysis revealed an n value of 1.25 for the CbzNaph-based device and 1.09 for the 2Ph-CbzNaph-based device, respectively, suggesting substantially attenuated Shockley–Read–Hall recombination at the 2Ph-CbzNaph/perovskite buried interface³⁶. The enhanced FF was considerably contributed from the reduced transport loss and non-radiative recombination. The improved charge dynamic behaviour also verified that there is faster transient photocurrent decay and slower transient photovoltage decay (Supplementary Fig. 46). The dark current–voltage tests represent less leakage current and enhanced shunt resistance when employed 2Ph-CbzNaph as the hole-selective layer (Supplementary Fig. 47). This shows the advantage of employing 2Ph-CbzNaph to provide better coverage on substrate with reduced charge carrier recombination.

Stability evaluation and mechanism analysis

The stability of PSCs is intimately linked to the intrinsic robustness of each functional layer and the structural integrity of their interfaces.

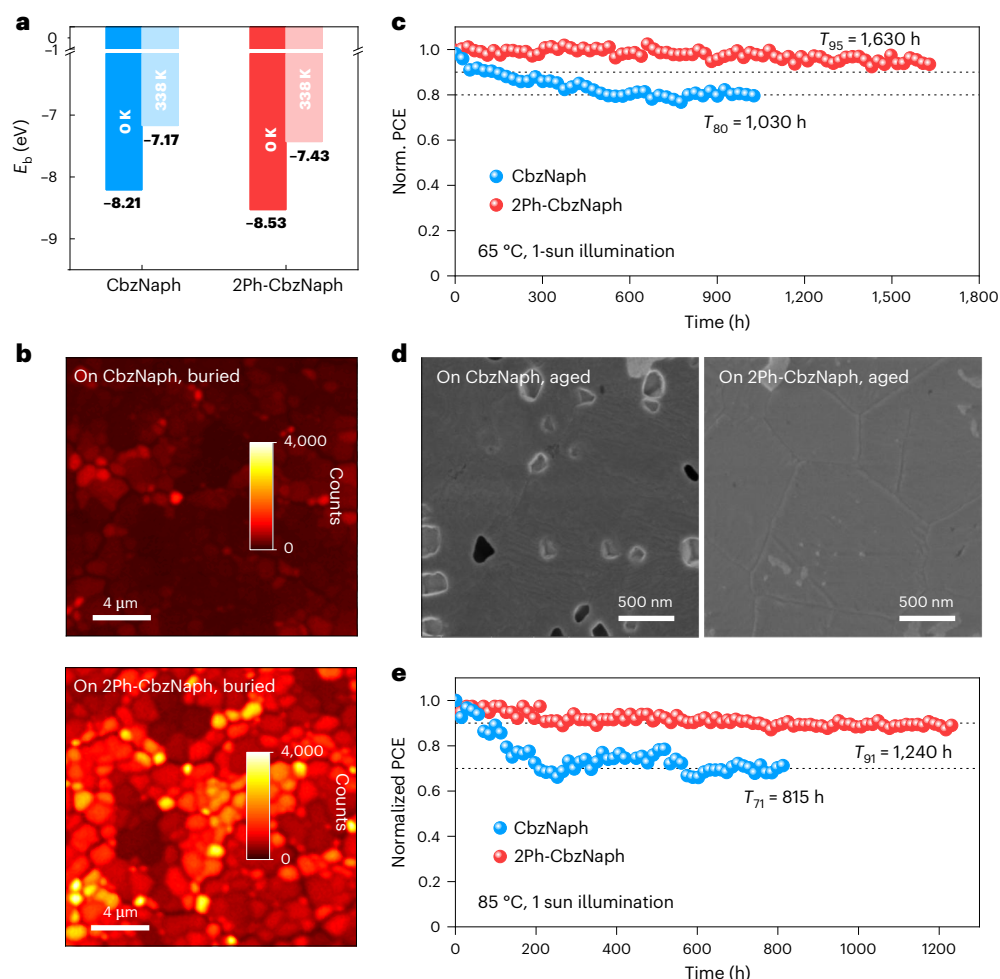


Fig. 4 | Device stability performances. **a**, Binding energies (E_b) of CbzNaph and 2Ph-CbzNaph with FTO substrates at the temperature of 0 K and 338 K. The y-axis break range is -1 to -6. **b**, PL intensity mapping of buried perovskite film peeled from SHM-covered substrates. **c**, Device operational stability comparison under 1-sun illumination at 65 °C; the initial efficiencies at the room temperature were

24.5% and 25.3% for CbzNaph and 2Ph-CbzNaph, respectively. **d**, Top-view SEM images taken from the buried interface of perovskite layers after ageing. **e**, Operational stability comparison under 1-sun illumination at 85 °C; the initial efficiencies at the room temperature were 24.4% and 25.2% for CbzNaph and 2Ph-CbzNaph, respectively.

Repeated CV scans were conducted to assess redox stability of the two SHMs. After 25 cycles, CbzNaph exhibited a notably decreased current intensity, whereas the one from 2Ph-CbzNaph still maintained relatively stable CV traces (Supplementary Fig. 48). This enhanced redox stability can be attributed to improved electron delocalization through two substituted phenyl group, which contributes to greater stability of 2Ph-CbzNaph.

To evaluate the stability of SHMs in devices under thermal stress, we further resorted to DFT calculation to evaluate their binding strength with the substrate at 0 K and 338 K (~ 65 °C) (Fig. 4a and Supplementary Figs. 29 and 49). 2Ph-CbzNaph exhibited a more negative binding energy (E_b) with FTO (-8.53 eV) compared to CbzNaph (-8.21 eV) at 0 K, anticipating a stronger substrate interaction. This scenario is in agreement with XPS results, which revealed more pronounced shifts in Sn 3d and O 1s peaks for 2Ph-CbzNaph (Supplementary Fig. 20e,f). While both their E_b decreased at 338 K, suggesting certain thermal sensitivity, 2Ph-CbzNaph maintained a relatively higher E_b than those of CbzNaph. It should be noted that both the energy and temperature of the SHM-SnO₂ systems converge quickly within the first 1.5 ps and then level off as a function of the simulation time, which confirms the SHMs reach an equilibrium state on the SnO₂ surface (Supplementary Fig. 50). These results point to superior thermal stability and interaction with a substrate for 2Ph-CbzNaph.

To evaluate the SHM/perovskite interface, related XPS measurements showed the shifted Pb 4f and I 3d peaks towards lower binding energies for the 2Ph-CbzNaph-based sample compared with CbzNaph, implying stronger interactions and interfacial stability (Supplementary Fig. 51).

We further examined the quality of perovskite crystallization at buried interface via GIWAXS. The 2D diffraction patterns showed enhanced intensity, indicating a promoted perovskite crystallization on the interface with 2Ph-CbzNaph (Supplementary Fig. 52). The corresponding integrated one-dimensional (1D) GIWAXS profiles confirmed higher crystalline for the perovskite film grown on 2Ph-CbzNaph compared to that on CbzNaph (Supplementary Fig. 53). Notably, a split peak was observed in the 1D profiles, probably originated from the interfacial strain or defect-induced lattice distortions³⁷. Spatially resolved photoluminescence (PL) imaging revealed higher PL intensity, suggesting reduced non-radiative trap states, due likely to the more homogeneous SHM assembly (Fig. 4b). We also collected the perovskite grain size at the buried layer from PL mapping results, the overall grain size grown on 2Ph-CbzNaph was around 1.6 μm , compared with 1.2 μm on CbzNaph (Supplementary Fig. 54). The larger grain size with fewer grain boundaries, could be helpful to reduce defect density and attenuate the non-radiative recombination pathways, which was correlated with higher PL intensity and ultimately enhanced device photovoltaic performance. In conclusion, 2Ph-CbzNaph provided a superior template

for facilitating the perovskite growth, which enhances the robustness of the interface against stress conditions under elevated temperature and light illumination.

Device performance under continuous operation was monitored following the ISOS-L-2 (International Summit on Organic Photovoltaic Stability, ISOS) protocol under 65 °C and 1-sun illumination³⁸ (Fig. 4c and Supplementary Figs. 55 and 56). The best 2Ph-CbzNaph-based device kept 95% of its initial efficiency (T_{95}) after being operated for 1,630 hours, whereas the CbzNaph-based devices reached T_{80} after 1,030 hours. To investigate the degradation pathway, aged devices were peeled off to expose their buried layers. The SEM images revealed substantial surface deterioration, including possible photoetching and obvious pinholes, in the CbzNaph-based device (Fig. 4d). This phenomenon may be associated with the suboptimal charge extraction of CbzNaph under operational stress, which promoted interfacial charge accumulation and ultimately accelerated perovskite decomposition. For higher temperature testing, we first performed a heat-only ageing test at 85 °C, the target device kept 94% of its initial PCE, compared to 89% for the control after 3,500 hours (Supplementary Fig. 57). Under the more demanding condition of combined 85 °C and 1-sun illumination, our best 2Ph-CbzNaph-based device maintained 91% of initial efficiency after 1,240 hours (Fig. 4e and Supplementary Fig. 58), whereas the CbzNaph-based device kept 71% of initial efficiency after 815 hours. These results confirm the superior thermal and operational stability imparted by 2Ph-CbzNaph (Supplementary Table 3). Moreover, the PCE loss for both devices were primarily driven by declines in FF and V_{oc} , underscoring the imperfect SHM/perovskite contacts and highlighting the need of further interface regulation in the future, as suggested by the $J-V$ characteristic results before and after ageing (Supplementary Fig. 59). What is more, we also conducted 1-cm² device comparison, as shown in Supplementary Fig. 60, after 335 hours continuous 1-sun illumination at 85 °C, our 1 cm² device based on CbzNaph and 2Ph-CbzNaph exhibited PCE retention rate of 67% and 96%, respectively.

Conclusions

We present a rational designed SHMs of 2Ph-CbzNaph, featuring a twisted molecular geometry induced by lateral π -conjugation of phenyl groups on the 7H-dibenzo[c,g]carbazole core. This design enables more intermolecular interactions that promote quasi-random oriented molecular assembly on conductive substrate. This is specifically important and beneficial as it facilitates smooth orientation transitions in SHM alignment from parallel at the perovskite interface to normal at the FTO interface, thereby enhancing the probability of optimal charge transfer. The resulting uniform and conformal interfacial morphology not only benefit charge-transfer kinetics but also support high-quality perovskite crystallization. These comprehensive findings establish 2Ph-CbzNaph as a superior benchmark SHMs, combining high efficiency, good scalability and stability, which are key milestones towards the commercial viability of perovskite photovoltaics. This work will further provide valuable insights for the exploration of more efficient hole-selective molecules.

Methods

Materials

The perovskite synthesizing materials include lead iodide (PbI₂), caesium iodide (CsI) and rubidium iodide (RbI) were purchased from Sigma-Aldrich. Formamidinium iodide (FAI) was purchased from Greatcell Solar. Methylammonium chloride (MACl), 3-(methylthio) propylamine hydroiodide (3MTPAI) were purchased from Xi'an Yuri Solar. Ethylenediammonium diiodide (EDAI₂) was purchased from Sigma-Aldrich. C₆₀ was purchased from Luminescence Technology Corp. Silver (purity 99.99%) was purchased from ZhongNuo Advanced Material (Beijing) Technology. The required solvents, including *N,N*-dimethylformamide (DMF, extra dry, purity 99.8%),

1-methyl-2-pyrrolidone (NMP, anhydrous, $\geq 99.9\%$), isopropanol (IPA, extra dry, purity 99.8%) were purchased from Sigma-Aldrich. Ethanol (EtOH, extra dry, 99.5%) was purchased from Thermo Scientific. All chemical reagents were used as received without further purification.

Precursor solutions

The composition of Cs_{0.025}Rb_{0.025}FA_{0.95}PbI₃ was adopted for this study. 1.8 M perovskite precursor solution was prepared by dissolving CsI (10.52 mg), RbI (8.6 mg), FAI (264.7 mg), PbI₂ (746.8 mg) in 745 μ l DMF and 155 μ l NMP. Additional 5% PbI₂ and 12.5% MACl was added to the precursor solution, and the solution was stirred for about two hours before use. The SHM solution was prepared by dissolving CbzNaph or 2Ph-CbzNaph in EtOH solvent with the concentration of 0.5 mg ml⁻¹.

Perovskite solar cells fabrication

The device structure was glass/FTO/SHMs/perovskite/modification layer/C₆₀/SnO₂/Ag. FTO glass was cleaned using detergent, deionized water, acetone and IPA before drying in a N₂ flow. Next, the FTO substrate was treated by ultraviolet ozone for 30 min and then transferred to a N₂ glovebox. A 0.5 mg ml⁻¹ SHM solution was deposited on the FTO at 2,500 rpm for 30 s and annealed at 100 °C for 10 min. The filtered perovskite precursor was spin coated on the substrates at 2,500 rpm for 12 s and quickly transferred to the vacuum chamber, underwent vacuum extraction (around 20 Pa) for 1 min. And then anneal the film at 100 °C for 30 min. After that, 1 mg ml⁻¹ EDAl₂ and 2 mg ml⁻¹ 3MTPAI in IPA was coated on the perovskite surface at 5,000 rpm for 25 s and annealed at 100 °C for 5 min. Fifteen nm C₆₀, 15 nm SnO₂ were finished by thermally evaporation and atomic layer deposition, respectively. Finally, 100 nm Ag was thermally evaporated (2.0 Å s⁻¹) as the electrode using a shadow mask. One hundred nm MgF₂ as an antireflective layer was thermally evaporated at the rate of 2.0 Å s⁻¹ when pushing the champion efficiencies. Note that when fabricating devices for stability tests, we did not use the anti-reflection layer for the devices stability testing and sputtered 8 nm indium tin oxide (ITO) as buffer layer between SnO₂ electron transport layer and electrode, and we used gold as our back electrode. The 1 cm² device was made on the 2 × 2 cm substrate and fabrication procedures were totally the same as the small device fabrications.

Device efficiency and stability tests

The device area was 0.108 cm², and a 0.0725 cm² black non-reflective mask was used for small-area device to define the accurate active cell area when conducting current density–voltage ($J-V$) measurements. For the 1-cm² device, the active area is 1.1 cm² and a 1 cm² mask was used. The efficiencies were recorded using a Keithley 2400 Source Meter under 100 mW cm⁻² (AM 1.5 G) illumination using a solar simulator (Enlitech, SS-X50R); a KG-5 Si photodiode was used to calibrate the solar simulator light intensity. The photovoltaic cells were measured both in reverse scan (1.2 V $\rightarrow$ 0 V, step 0.02 V, dwell time 100 ms) and forward scan (0 V $\rightarrow$ 1.2 V, step 0.02 V, dwell time 100 ms) in a nitrogen glovebox, under room temperature. For the long-term stability tracking, the illumination was provided by a light-emitting diode (LED)-based solar simulator, calibrated to approximately 1-sun intensity. The spectrum was provided in Supplementary Fig. 61. Samples were put into a tightly closed chamber. N₂ flow was maintained during stability monitoring. And throughout the operational tracking duration, a hot plate served as the thermal source to maintain the device surface temperature at approximately 65 °C and 85 °C, a sensor was inside the chamber to monitor the actual temperature. The steady-state power output was monitored by applying a constant bias voltage of 800 mV, the testing system was supported by Wuhan 91PVKSolar Technology. Throughout the operational process, $J-V$ curves were also periodically recorded.

Material, film and device characterizations

Thin film X-ray diffraction measurements were conducted using a D8 ADVANCE diffractometer (Bruker) with Cu K α radiation ($\lambda = 0.15418$ nm)

at 1,600 W power (40 kV, 40 mA). The GIWAXS measurements were performed at the BL02U2 and BL17B1 beamlines of Shanghai Synchrotron Radiation Facility by using X-ray with a wavelength of 1.2398 Å. SHM films were prepared by spin coating a 2 mg ml⁻¹ solution in EtOH. Both SHM and perovskite samples were measured at an incidence angle of 0.20° relative to the substrate plane. The acquired data were analysed using the Dioptas software.

XPS and UPS spectra were acquired using an ESCALAB 250Xi spectrometer (Thermo Fisher). For XPS measurements, the X-ray source was Al Kα (1,486.6 eV) of 150 W. The photoelectron pass energy was 100 and 20 eV for the survey and narrow XPS spectra, respectively. The Thermo Scientific Advantage software was used for data acquisition and peak fitting. The binding energy was calibrated against the C 1s peak at 284.8 eV before analysis. UPS measurements were performed using a He I radiation with an excitation energy of 21.22 eV and a hemispherical energy analyser. The UPS spectra were collected under a pressure of 2.3 × 10⁻⁸ mbar with a pass energy of 2 eV and a negative bias of -10 V. The spectrometer Fermi level was calibrated with a standard gold sample.

The cyclic voltammetry experiments were performed on a CHI1040C electrochemical workstation (CH Instruments) in conjunction with a three-electrode system at room temperature in a nitrogen atmosphere. The SHM-coated FTO served as the working electrode, with Pt wire serving as the counter electrode and an Ag/AgCl (saturated KCl) acted as the reference electrode. The electrolyte consisted of 0.1 M tetrabutylammonium phosphorus hexafluoride (Bu₄NPF₆) in 1,2-dichlorobenzene (*o*-DCB) and a series of potential scan rates spinning from 0.025 V s⁻¹ to 0.3 V s⁻¹ was applied. The surface density of SHM molecules on FTO can be determined from cyclic voltammetry using the equation below³⁹:

$$i_p = \frac{n^2 F^2}{4RTN_A} A \Gamma^* \nu$$

where i_p is the oxidative peak current, ν is the potential scan rate (V s⁻¹), n is the number of electrons transferred, the Faraday constant $F = 96485.33$ C mol⁻¹, the universal gas constant $R = 8.314$ J mol⁻¹ K⁻¹, T is the temperature, the Avogadro constant $N_A = 6.022 \times 10^{23}$ mol⁻¹, the electrode surface area $A = 1.5$ cm² in this work and Γ^* (molecules cm⁻²) is the surface density that can be derived via the slope of i_p versus ν plots.

Kelvin probe force microscopy (KPFM) and conductive atomic force microscopy (C-AFM) measurements were performed with the Multimode8 microscope (Bruker) in tapping and contact mode, respectively, utilizing a HQ:NSC18/Co-Cr tip (MikroMasch). For C-AFM measurements, a bias of 400 mV was applied.

For the space-charge-limited current measurement to determine the hole mobility, we fabricated hole-only devices with an architecture: ITO/PEDOT:PSS/SHMs/MoO₃/Ag. Dark current density–voltage (J – V) curves based on the hole-only devices were recorded under a potential bias range from 0 to 5 V, employing a Keithley 2400 source meter. The hole mobility (μ) was derived via fitting J – V curves by using the Mott–Gurney law^{40,41}:

$$J = \frac{9}{8} \varepsilon_0 \varepsilon_r \mu \frac{V^2}{L^3}$$

where J is the hole current density, V is the applied voltage, ε_0 and ε_r is the vacuum and relative permittivity, respectively, and L is the thickness of the SHM layer. We used ε_r of 3 as normally for organic small molecules and the thickness of the SHM layer (~30 nm) was confirmed via step profile.

For the conductivity measurement, the structure of ITO/SHMs/Au was used and conducted the J – V measurements, the direct-current conductivity (σ) can be obtained according to the formula of $J = \sigma V/L$.

Top-view and cross-sectional Scanning electron microscopy images were recorded on a Nova NanoSEM 650 field emission scanning electron microscope (Thermo Fisher Scientific) at an accelerating voltage of 2 kV.

Transient reflectance spectroscopy was measured through the reflection modes of the pump-probe system. The fundamental beam is an 800-nm laser generated by a Ti:sapphire amplifier with a repetition rate of 1 kHz. This fundamental pulse is split into two branches by a beam splitter. One branch passes through an optical parametric amplifier (TOPAS) to generate the pump pulse (290 nm–2,600 nm). The pump pulse is modulated at a frequency of 500 Hz and attenuated by a neutral density filter wheel. The other 800 nm branch is focused into a sapphire crystal to generate a white-light continuum spanning (350 nm–1,600 nm). A motorized translation stage with a retro-reflecting mirror is used to achieve the time delay between the pump pulse and the probe pulse (up to 8 ns). The pump and probe beams are spatially overlapped on the sample surface. The probe beam incident angle is fixed at 45°, while the pump beam incidents the sample normally. The focused spot size at the sample position is approximately 200 μm for the probe beam and 600 μm for the pump beam.

To quantitatively describe the carrier density evolution under the various optical excitation conditions, a one-dimensional diffusion model shown below was used

$$\frac{\partial N(x, t)}{\partial t} = D \frac{\partial^2 N(x, t)}{\partial x^2} - \frac{N(x, t)}{\tau_B}$$

where $N(x, t)$ represents the carrier density as a function of film depths (x) over times (t), D is the ambipolar diffusion coefficient and τ_B is the body carrier lifetime.

By assuming that the initial carrier generation happens instantaneously and within the time resolution of the pump excitation pulse, the initial condition is given by the equation:

$$N(x, t)|_{t=0} = N_0 e^{-\alpha x}$$

where N_0 is the initial surface carrier density, such that $N_0 = \alpha(1 - R)J_0$, α is the absorption coefficient at the selected photon energy, R is the reflectance and J_0 is the pump fluence.

For the polycrystalline samples, the boundary conditions are described as

$$\begin{aligned} \left. \frac{\partial N(x, t)}{\partial t} \right|_{x=0} &= \frac{SEV}{D} N(0, t) \\ \left. \frac{\partial N(x, t)}{\partial t} \right|_{x=L} &= -\frac{SRV}{D} N(L, t) \end{aligned}$$

where L is the film thickness. Generally speaking, in metal halide perovskites, the bulk recombination can be neglected under a time delay of 7 ns, so the bulk recombination term is ignored. When the molecules are present in the sample, the boundary condition is set as the extraction from the front surface being dominant, and this process occurs concurrently with the recombination of the back surface. By using the numerical solution $N(0, t)$, the transient reflection (TR) kinetic traces at a suite of wavelengths (Fig. 2e,f) are modelled with a global fit method which utilizes the above diffusion model and boundary conditions to obtain the optimal fitting values of surface extraction velocity. In the same perovskite material system, the diffusion coefficient D and the surface recombination rate are inherent properties of the material. Therefore, the surface recombination rate of single-layer perovskite films can be selected as the parameter value of post-surface recombination velocity, being $1,080 \pm 148$ cm s⁻¹. The D value for the single-layer perovskite film was determined to be 2.09 ± 0.12 cm² s⁻¹, which was set as a constant in this study.

Time-resolved photoluminescence (TRPL) measurements were executed on an FLS1000 fluorescence spectrometer (Edinburgh). An EPL-375 picosecond laser diode was used to supply excitation light. The laser pulse fluence is around 4 nJ cm⁻². The repetition rate is 0.2 MHz. When employed fitting techniques on the TRPL transients to calculate the differential time (τ), defined by the equation of $\tau = -\{d\ln[\varphi(t)/dt]\}^{-1}$, where $\varphi(t)$ represents the time-dependent PL photon flux. PL Mapping was measured by using the Micro Time 100 confocal microscopic system (PicoQuant). An Olympus 100× oil immersion objective with

a numerical aperture of 1.45 was used for the measurement. Samples were covered by immersion oil (Olympus Immersion Oil Type-F) in a dark environment during the measurement. The picosecond 640-nm pulsed laser with a repetition rate of 5 MHz was used to excite the perovskite films. The average laser intensity was kept at around 12 nW at the repetition rate of 5 MHz, corresponding to a photon energy of ~ 2.4 fJ per pulse. The PL generated by samples was collected through the same objective and relayed onto an image plane through a telescopic relay lens system. The transmitted epifluorescence image was then collected by a single photon detector (PicoQuant PMA Hybrid) through an optical fibre that was connected to a precise x-y piezo motor stage (PI U-751.24) with a linear step resolution of 100 nm.

For dark I - V measurement, the voltage continuously sweeps from -1.0 V to 1.5 V with a step of 0.05 V, using a Keithley 6482 Source Meter. Transient photocurrent and photovoltage were obtained using the all-in-one characterization platform PAIOS (Fluxim AG).

Dynamic light scattering measurements were performed on a Dybapro NanoStar (Wyatt Technology) particle size analyser. X-ray reflectivity measurements were conducted on a D8 DISCOVER diffractometer (Bruker) with Cu K α radiation at $1,600$ W power (40 kV, 40 mA). The data were fitted by using the LEPTOS software via the Genetic Algorithm method. In Supplementary Figs. 13 and 14, the single crystal data were visualized by Olex2 software⁴².

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available within the article and its Supplementary Information. Source data are provided with this paper.

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Author contributions

Qi Jiang, T.L., W.J. and T.W. conceived the idea. Qi Jiang supervised the project and process. T.L. fabricated the perovskite films and devices and conducted most of the measurements and characterizations. W.J. synthesized and characterized 2Ph-CbzNaph and further analysis. T.W. and Q.D. conducted the theoretical calculations and related analysis. Y.Y. performed GIWAXS measurements and analysis. J.L. conducted TR measurements and analysis under the supervision of Y.Z. C.L. conducted ARPR and analysis under the supervision of M.L. and P.T. Z.J. and X.W. conducted SFG-VS and analysis. H.Z. and Z.N. conducted PL mapping and analysis. Qi Jiang and T.L. wrote the paper. A.K.-Y.J. finalized the paper. All authors were involved in the discussion during the project.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Alex K.-Y. Jen or Qi Jiang.

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¹State Key Laboratory of Semiconductor Physics and Chip Technologies, Institute of Semiconductors, Chinese Academy of Sciences, Beijing, China.

²Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing, China. ³Department of Materials Science and Engineering, City University of Hong Kong, Hong Kong, China. ⁴Key Laboratory of Automobile Materials, Ministry of Education, and School of Materials Science and Engineering, Jilin University, Changchun, China. ⁵School of Microelectronics, Fudan University, Shanghai, China. ⁶Key Laboratory of Low-Dimensional Quantum Structures and Quantum Control of Ministry of Education, Department of Physics, Hunan Normal University, Changsha, China. ⁷School of Chemistry and Chemical Engineering, Key Laboratory of Surface and Interface Science of Polymer Materials of Zhejiang Province, Zhejiang Sci-Tech University, Hangzhou, China. ⁸State Key Laboratory of Silicon and Advanced Semiconductor Materials and School of Materials Science and Engineering, Zhejiang University, Hangzhou, China. ⁹Key Laboratory of Bionic Engineering, Ministry of Education, Jilin University, Changchun, China.

¹⁰These authors contributed equally: Tianyu Li, Wenlin Jiang, Tonghui Wang. ✉e-mail: alexjen@cityu.edu.hk; qjiang@semi.ac.cn

Solar Cells Reporting Summary

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells

☒ Yes
☐ No

For small area solar cells, the active area is 0.108 cm² and mask size is 0.0725 cm²; for 1 cm² device, the active area is 1.1 cm² and mask size is 1 cm².

Explain why this information is not reported/not relevant.

Method used to determine the device area

☒ Yes
☐ No

The device areas were defined by a black non-reflective metal mask.

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction

☒ Yes
☐ No

All the J-V curves presented in the Main manuscript and Supplementary Information all shown in both forward and backward direction measurement results.

Voltage scan conditions

☒ Yes
☐ No

Please find in the Method section. The cells were measured 1.2 V → -0.1 V for backward and -0.1 V → 1.2 V for forward, step size 0.02 V, dwell time 100 ms.

Explain why this information is not reported/not relevant.

Test environment

☒ Yes
☐ No

Please find in the Method section. The devices were measured in nitrogen glovebox, under room temperature.

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization

☐ Yes
☒ No

Provide a description of the protocol.

No preconditioning before testing.

Stability of the J-V characteristic

☒ Yes
☐ No

The stable power output (SPO, 300s or 600 s) or maximum power point (MPP, certification, 300 s) tracking data, please refer to Fig. 3b and 3e, Supplementary Figs. 40, 42, 43b, they are consistent with the J-V scans measurement results.

Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization

☐ Yes
☒ No

Provide a description of hysteresis or any other unusual behaviour observed during the characterization.

No hysteresis or any other unusual behaviour observed.

Related experimental data

☐ Yes
☒ No

Provide a description of the related experimental data.

No hysteresis or any other unusual behaviour observed.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)

☒ Yes
☐ No

Supplementary Fig. 38, 43c; EQE were measured using an integrated system, from Enlitech

Explain why this information is not reported/not relevant.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator

☒ Yes
☐ No

The integrated J_{sc} from EQE spectra is consistent with those from J-V testing (all less than 3% error)

Explain why this information is not reported/not relevant.

For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	Provide a description of the measurement conditions. No tandem device presented in this work.
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	The light source is from Enlitech. Co., Ltd., SS-X50R, a KG-5 Si photodiode was used to calibrate the solar simulator light intensity to 100 mW/cm ² AM 1.5G. Explain why this information is not reported/not relevant.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	The standard Si reference cell was purchased from Enli Tech. Co., Ltd., Taiwan, which was calibrated and certified by NREL. Explain why this information is not reported/not relevant.
Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	The correction spectral mismatch factor (MFF) was used in the test software, the MFF is around 0.99-1.008. Explain why this information is not reported/not relevant.
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	0.0725 cm ² and 1 cm ² mask were used for small area and 1 cm ² device. Explain why this information is not reported/not relevant.
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	Report the difference in the short-circuit current density values measured with the mask and aperture area. No obvious variation.
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	The small area and 1 cm ² device were certified by National PV Industry Measurement and Testing Center (NVTM). Explain why this information is not reported/not relevant.
A copy of any certificate(s)	☒ Yes ☐ No	Supplementary Fig. 40, 42. The small area device was certified with a 26.67% stabilized efficiency, the 1 cm ² device was certified with a 25.94% stabilized efficiency. Explain why this information is not reported/not relevant.
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	30 reference cell and 50 target cell were tested. Explain why this information is not reported/not relevant.
Statistical analysis of the device performance	☒ Yes ☐ No	Fig. 3c, Supplementary Fig. 42. Targeted device exhibited better average yields and narrower distributions in all FF, Voc and overall PCE. Explain why this information is not reported/not relevant.
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	Operational tracking stability: devices were unencapsulated and put into a tightly closed chamber, nitrogen flow was maintained during testing and a hot plate was served as the thermal source for 65°C and 85°C, and a sensor was inside the chamber to monitor the actual temperature; a constant bias voltage of 800 mV was applied, the light intensity was calibrated to 1-sun and the illumination spectrum can be found in Supplementary Fig. 60; Thermal stability: unencapsulated, 85°C, dark in nitrogen-filled glovebox. Explain why this information is not reported/not relevant.